The Augustine Problem in Agents

Aojie (Justin) Yuan
University of Southern California
aojieyua@usc.edu

Yue Zhao
University of Southern California
yzhao@usc.edu

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Abstract

Large language model agents are trained on token sequences and deployed in wall-clock time. They cannot perceive the wall-clock, and we prove this perception cannot be acquired from any token-only training loss. The Chronoception Impossibility Theorem (CIT) makes the claim precise: the gradient of any loss whose support lies entirely in token sequences carries no signal aligning external wall-clock time with any internal representation. The Three Times ontology — τ_{wall} (external), τ_{step} (policy invocations), τ_{self} (the agent's narration) — names the projections an agent's policy must align; the *Augustine Problem* is the failure of that alignment. ChronoBench, a 9-capability diagnostic suite, measures the failure on 10 agent configurations across 3 providers ($\sim$4,000 trajectories).

Five findings emerge. **(i)** Every panel agent honors less than 5% of the wall-clock budget given to it (L2 Step-Clock Conflation), and date injection does not move the number ($|\Delta\text{CAR}| \leq 0.01$). **(ii)** Self-narration error $|\rho|$ closes monotonically across five non-reasoning frontier generations ($+1.12 \to +0.07$, a 94% reduction) while CAR does not change: the narrative axis is text-trainable; the action axis is not. **(iii)** The *Reverse-Scaling Theorem* (Theorem 2): under CIT, reasoning-token expansion monotonically grows $|\rho|$ within a fixed agent. We confirm it intra-model (o4-mini at three reasoning-effort levels: $|\rho|$ goes $1.23 \to 1.54 \to 1.67$) and cross-model (Claude Sonnet 4.6 with versus without extended thinking: $0.07 \to 0.16$, the sign of ρ flipping). **(iv)** The *Calibration Catastrophe*: a nominally 90% confidence interval on the agent's own work-duration achieves at most 50% actual coverage across the panel; GPT-4o achieves 0%. **(v)** The *Injection Tell*: every consumer web-chat product we audited injects today's date into its system prompt (3/3, verbatim leaked-prompt evidence); no developer-tool product does (0/3). Competing labs converge on the same workaround independently — a natural experiment that the underlying models lack a representation of time.

We argue these are not five surprises but one consequence. Aggregated as a single scalar ε, the failures place every panel agent above the Augustine threshold $\varepsilon^* = 0.20$ (calibrated at $2\times$ human introspective error). The action-axis component alone exceeds the threshold for every agent: narrative-axis training cannot close the gap, and reasoning-token expansion widens it. The Agentic Timeline Hypothesis turns the measurement into a deployment bound, $T_{\max}(A) \propto 1/\varepsilon(A)$. Pre-registered prediction P12 is qualitatively supported by 14,709 public METR HCAST runs: reasoning models decay 20% faster than non-reasoning models at matched short-horizon success. The framework generalises along its natural other axis: a Spatiotemporal Impossibility Theorem (SIT, Theorem 3), the *Cartographic Problem*, and the *Agentic Frontier* bound $T_{\max} \cdot S_{\max} \leq C/\varepsilon_{ST}$ extend the claim to the joint (T, S) plane on which long-horizon agents are deployed — the bridge to a planned third paper. We release ChronoBench, the Injection Atlas, all $\sim$4,000 trajectories, a Docker reproducibility package, and a locked OSF pre-registration.

1 Introduction

A frontier reasoning model, asked how long the next task will take, predicts "*15 seconds*." It thinks for ninety. It answers in two. Pressed afterwards on the duration, the same model reports "*0.04 seconds*." Pressed for a 90% confidence interval, the interval contains the truth 0% of the time. None of these failures is a bug or a misread of a particular prompt. They are joint, panel-wide, and structurally predictable. We measure them across ten frontier agent configurations from three providers and roughly four thousand trajectories; the joint occurrence is present in every model we tested.

The Augustine Problem. The five training losses in current practice — next-token cross-entropy, supervised fine-tuning, RLHF, RL with verifiable rewards, reasoning supervision — are all functionals of token sequences. None of them contains wall-clock time in its support. Yet the same models are deployed as agents that act in continuous physical time, where budgets are measured in seconds and consequences unfold without regard to step count. We call the resulting representational gap *the Augustine Problem*, after Augustine's confession that he knew what time was until asked to explain it.[1] The gap is structural, not engineering: we prove (Theorem 1) that under any token-only loss the policy gradient contains no signal aligning the wall-clock with any internal representation. Chronoception cannot be acquired from token-only training. It must be installed.

Three Times. Any agent trajectory admits three ontologically distinct projections of time: wall-clock time τ_{wall} (continuous, external), step time τ_{step} (discrete, the policy's invocation count), and self-narrated time τ_{self} (the agent's report of its own duration). A chronoceptively grounded policy enforces the implicit identity $\tau_{\text{wall}} \approx \tau_{\text{step}} \cdot \langle \Delta t \rangle \approx \tau_{\text{self}}$. The Augustine Problem is precisely the failure of a policy to enforce this identity — a property measurable on single-turn tasks, before any long-horizon dynamics can compound.

What goes wrong, and how. Three named empirical laws structure the panel's behaviour. *L1 Agentic Parkinson's Law* (coefficient α): budget-aware-trained agents fill the wall-clock budget they are given; native untrained agents do not. *L2 Step-Clock Conflation* (Clock-Adherence Ratio CAR): under wall-clock budgets, every panel agent uses less than 5% of the budget — agents silently degrade wall-clock budgets into step-count terminators. *L3 Temporal Confabulation* (ratio $\rho = \log_{10}(\tau_{\text{self}}/\tau_{\text{wall}})$): agents misreport the duration of their own work, with rich internal structure detailed in §6. The three laws aggregate into a single scalar ε.

Five findings, one consequence. **(i)** Every panel agent honors less than 5% of the wall-clock budget given to it; date injection does not move the number. **(ii)** Self-narration error closes along the non-reasoning capability axis ($|\rho|$ falls 94% across five frontier generations) but CAR does not. The narrative axis is text-trainable; the action axis is not. **(iii)** The Reverse-Scaling Theorem (Theorem 2): reasoning-token expansion monotonically grows $|\rho|$ within a fixed agent. The frontier strategy currently driving capability scaling — longer chains of thought, larger thinking budgets, more test-time compute — structurally degrades chronoception on the narrative axis. We confirm the theorem intra-model and cross-model, with sign-flipping under extended thinking. **(iv)** The Calibration Catastrophe: every panel agent's nominally 90% interval on self-duration

[1] We use *chronoception* in the cognitive-science sense [Wittmann, 2009, Eagleman, 2008]: the perception of one's own work duration. This is distinct from Goel et al. [2025]'s use of the same term for the temporal validity of facts in retrieval-augmented generation.

achieves at most 50% coverage; GPT-4o achieves zero. **(v)** The Injection Tell: consumer web-chat products inject today's date into their system prompts at 100% rate; developer-tool products at 0%. Competing labs converge on the same workaround independently, with substantial commercial incentive to ship a cheaper alternative if one existed. We argue this is decisive non-experimental evidence that the underlying foundation models lack a representation of time.

The five findings are not five surprises. They are projections of one structural fact: the action axis cannot be closed by narrative training, and reasoning-token expansion makes the narrative axis worse, not better. The Augustine Threshold $\varepsilon^* = 0.20$ (calibrated at 2× human introspective error) is crossed by no panel agent. Sonnet 4.6 — the best chronoceptive agent we measured — sits at $\varepsilon = 0.32$, with the action axis alone contributing 0.32. There is no narrative-axis intervention that closes the gap.

From measurement to deployment. Why does any of this matter? Industry is moving from minute-scale chat to hour-scale coding agents to day-scale autonomous research agents. At each new horizon a tighter chronoceptive calibration is required to honour budgets, plan against deadlines, and report progress. The Agentic Timeline Hypothesis (§9) makes this precise: each agent has a maximum viable deployment horizon $T_{\max}(A)$ that scales as $1/\varepsilon(A)$, modulated by CAR. Because L2 does not scale with capability and the Reverse-Scaling Theorem makes the narrative axis worse with reasoning compute, $T_{\max}$ scales with neither. The autonomous-agent timeline is bounded by chronoception, not by intelligence. Pre-registered prediction P12 is qualitatively supported on 14,709 public METR HCAST runs: reasoning models decay 20% faster than non-reasoning models at matched short-horizon success.

From time to space. The framework develops along one axis. But every long-horizon agent benchmark on which deployment claims are made — SWE-Bench, WebArena, GAIA, MLE-Bench — requires the agent to navigate space as well as time: a codebase, a website, the open web. The argument that produces CIT goes through unchanged on any external metric the loss does not see. The spatial face of the Augustine Problem is the *Cartographic Problem*, formalised in §12 as Theorem 3 (the Spatiotemporal Impossibility Theorem) with three spatial laws (SL1/SL2/SL3) mirroring L1/L2/L3. The temporal Agentic Timeline becomes the joint Agentic Frontier $T_{\max}(A) \cdot S_{\max}(A) \leq C/\varepsilon_{ST}(A)$. The framework's structural claim is that the autonomous-agent capability curve saturates not because intelligence saturates, but because spatiotemporal cognition does not scale. This is the bridge to a planned Paper 3.

Contributions. The Augustine Problem as a named structural diagnosis. The Three Times ontology with a measurable identity. The Chronoception Impossibility Theorem (Theorem 1) and its empirical corollaries. The **Reverse-Scaling Theorem** (Theorem 2) and its dual confirmation. The **Calibration Catastrophe** (§6.3). A **toy positive control** that crosses the Augustine threshold on T3.1 via wall-clock-supported LoRA SFT (§6.4), demonstrating CIT's constructive converse. ChronoBench (9 sub-capabilities, 10-agent panel, ∼4,000 trajectories), the Closed-Lab Injection Atlas (11 harnesses), and the **Agentic Timeline Hypothesis** with HCAST evidence (§9). The spatiotemporal generalisation: Theorem 3, the Cartographic Problem, and the Agentic Frontier (§12). A Docker reproducibility package, an Operational Chronoception Audit Checklist (Appendix F), and a locked OSF pre-registration. All benchmark code, trajectories, and analysis scripts are released.

2 The Three Times

To name the gap, we need a vocabulary for what it is a gap in. We decompose any agent trajectory into three ontologically distinct projections of time (Figure 1).

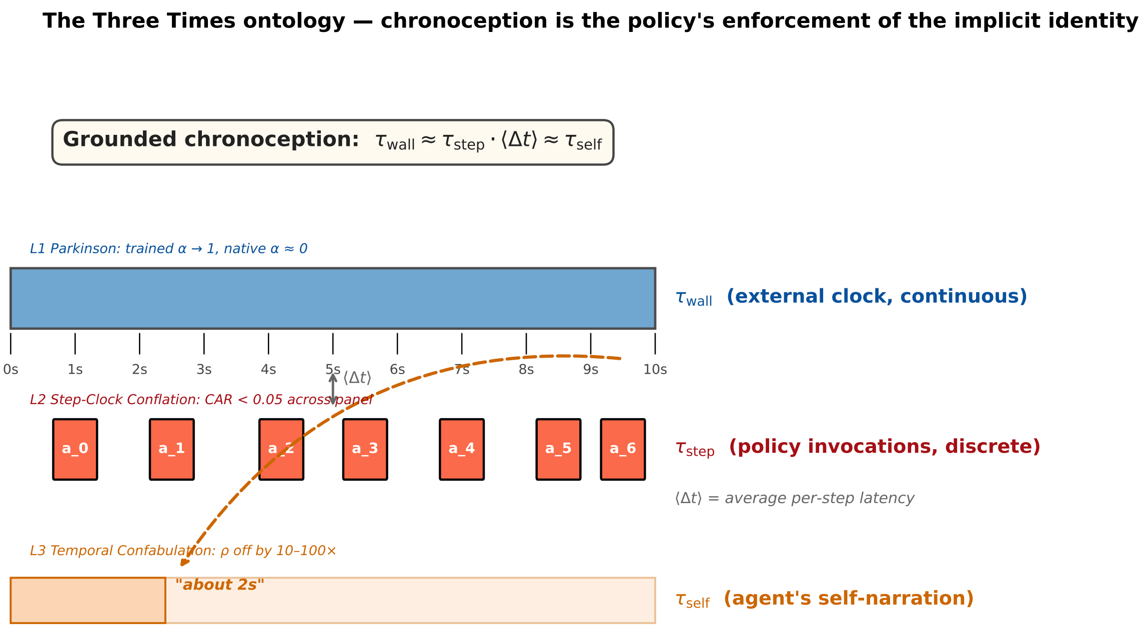

Figure 1: The Three Times. τ_{wall} (top, blue) is external and continuous. τ_{step} (middle, red) is the discrete sequence of policy invocations $a_0, a_1, \ldots$. τ_{self} (bottom, orange) is the agent's narrative of its own duration. A chronoceptively grounded policy enforces $\tau_{\text{wall}} \approx \tau_{\text{step}} \cdot \langle \Delta t \rangle \approx \tau_{\text{self}}$; the Augustine Problem (dashed arrow) is the failure of this identity. Each axis carries a named empirical law (L1, L2, L3).

Definition 1 (Three Times)**.** *For an agent A executing trajectory $\tau = (s_0, a_0, s_1, a_1, \ldots, s_n)$:*

- ***Wall-clock time*** $\tau_{\text{wall}}(\tau) := t(s_n) - t(s_0)$ *assigns each state an external timestamp.*
- ***Step time*** $\tau_{\text{step}}(\tau) := n$ *counts policy invocations (LLM calls or sub-trajectories).*
- ***Self-narrated time*** $\tau_{\text{self}}(\tau) := \Pi(s_n^{output})$ *extracts the agent's reported duration from its terminal output. The parser is specified in Appendix A.*

Grounded chronoception requires the implicit identity

$$\tau_{\text{wall}} \approx \tau_{\text{step}} \cdot \langle \Delta t \rangle \approx \tau_{\text{self}},$$

where $\langle \Delta t \rangle$ is the agent's average per-step wall-clock cost. The first approximation is L2 alignment: step counts track wall-clock. The second is L3 alignment: the agent's narrative tracks reality. The Augustine Problem is what we observe when both fail.

A natural objection: surely τ_{step} is just τ_{wall} in different units? Empirically, no. Under wall-clock budgets, the policy uses step count as its binding terminator (§5); under step budgets, the same policy uses step count. The agent's internal control variable is τ_{step}, and the gap $|\tau_{\text{wall}} - \tau_{\text{step}} \cdot \langle \Delta t \rangle|$ is precisely what we measure. We use *chronoception* in the cognitive-science sense (perception of one's own work duration), distinct from Goel et al. [2025]'s use of the same term for fact validity, and from Chu et al. [2024]'s temporal reasoning over external timelines.

With the vocabulary fixed, the Augustine Problem can be stated precisely; we do so next.

3 The Augustine Problem

Definition 2 (Augustine Problem)**.** *Agent A exhibits the* Augustine Problem *on task family $\mathcal{T}$ if its policy fails to enforce the identity of Definition 1:*

$$\mathbb{E}_{\tau \sim \pi_A} \big[\, |\tau_{\text{wall}} - \tau_{\text{step}} \langle \Delta t \rangle| + |\tau_{\text{wall}} - \tau_{\text{self}}| \,\big] > \varepsilon_0,$$

for some calibration tolerance ε_0. The agent's three projections of time drift independently across its trajectory distribution.

The question we attack first is not whether agents *do* exhibit the Augustine Problem (§5–6 show they do) but whether any conceivable foundation-model training procedure under current practice could in principle close it. We argue: not by additional scale, not by reasoning-compute expansion, not by any post-training intervention that operates on token sequences alone.

Theorem 1 (Chronoception Impossibility, CIT)**.** *Let $\mathcal{L}(\theta) = \mathbb{E}_{x \sim \mathcal{D}} \big[\, \ell(f_\theta(x_{<i}), x_i) \,\big]$ be a training loss where $\mathcal{D}$ is a distribution over token sequences and neither ℓ nor $\mathcal{D}$ contains wall-clock support. Then $\nabla_\theta \mathcal{L}$ contains zero gradient signal aligning τ_{wall} with any internal representation of A. Equivalently, the resulting policy enforces no implicit identity between τ_{wall} and either τ_{step} or τ_{self}.*

Sketch. $\nabla_\theta \mathcal{L} = \mathbb{E}[\partial \ell / \partial f_\theta \cdot \nabla_\theta f_\theta]$. Neither factor is a function of the wall-clock time at which the sample was generated: tokens carry no timestamps. The gradient is therefore invariant to arbitrary reparameterisation of τ_{wall} — including the scaling that maps one second to one hour. A policy obtained from this gradient cannot distinguish one wall-clock duration from another. □

What CIT covers. The losses in current practice — next-token cross-entropy, supervised fine-tuning, RLHF, RL with verifiable rewards, reasoning supervision — are all functionals of token sequences alone, and therefore all satisfy CIT's assumption. The theorem does *not* preclude harness-installed chronoception (the Injection Tell, §7, is precisely this kind of installation); nor does it preclude future losses that take wall-clock as an argument (the constructive program of ChronoStack, Paper 2). It precludes only the inference — which we believe is widely held but rarely defended — that additional scale, longer reasoning, or larger context will eventually close the gap. They will not, because they do not change the loss.

The empirical predictions of CIT. Three predictions follow directly. **(P-CIT-1)** Under wall-clock budgets, the policy uses step count as terminator: CAR stays bounded away from 1 regardless of capability (§5). **(P-CIT-2)** Prompt-level wall-clock information cannot move action-axis behaviour: $|\Delta\text{CAR}|$ is small under Setting B (§5, §7). **(P-CIT-3)** Calibration tooling (temperature scaling, isotonic regression, RLHF-with-confidence) cannot close calibration on self-duration, because the calibration target is not in the loss support (§6.3). All three are confirmed by the panel.

The Augustine Threshold. We adopt $\varepsilon^* = 0.20$ as the operational threshold for chronoceptive grounding, calibrated against the human introspective baseline $\varepsilon \approx 0.10$ [Wittmann, 2009] at twice the human error. The threshold is not the framework's central claim; it is a measurement target. No panel agent crosses it (§8). The interesting question, addressed in the rest of the paper, is which axis is responsible for the gap.

The benchmark we use to measure that is described next.

4 ChronoBench

ChronoBench is the instrument with which we test the predictions of CIT. The design philosophy is single-axis isolation: every instance loads on exactly one of the three Times, so a failure is unambiguously attributable to the failure mode it targets. We summarise the 9 sub-capabilities here; the full prompt set, panel composition, parser specification, and open-source serving recipe appear in Appendix A.

Table 1: The 9 sub-capabilities of ChronoBench, three per axis. Each row isolates a single chronoceptive failure mode for separate measurement.

Axis	Sub-capability	Tests
τ_{wall}	T1.1 Clock awareness	Today's date or current time
	T1.2 Elapsed-time tracking	Time since conversation start
	T1.3 Deadline-aware tradeoff	Strategy under wall-clock deadline
τ_{step}	T2.1 Step-budget honoring	Produce exactly N reasoning steps
	T2.2 Step-to-wall translation	Compute $N\times$ latency = duration
	T2.3 Wall-budget execution	Honour a wall-clock budget (L2 core test)
τ_{self}	T3.1 Retrospective ρ	Duration reported after task completion
	T3.2 Prospective ρ	Duration predicted before task begins
	T3.3 Calibration	90% CI over self-duration

Settings. Every instance runs under two settings. Setting A uses a vanilla system prompt ("*You are a helpful assistant.*") and supplies no harness-side wall-clock signal. Setting B prepends "*Current date and time: {ISO}*" to the system prompt, emulating the consumer web-chat behaviour documented in §7. The contrast between A and B isolates injected information from policy-internal representation.

Panel. Ten agent configurations from three providers: OpenAI's gpt-4o-mini, gpt-4o, gpt-5.1, o3, and o4-mini (three `reasoning_effort` levels: low, medium, high); Anthropic's Claude Haiku 4.5 and Claude Sonnet 4.6 (with and without extended thinking, 8k-token budget); open-source Qwen2.5-7B and DeepSeek-R1-Distill-Qwen-14B (three `max_completion_tokens` budgets), both self-hosted via vLLM on a single 8×RTX 6000 Ada cluster. The o4-mini reasoning-effort sweep and the Sonnet 4.6 thinking variant supply the empirical basis for the Reverse-Scaling Theorem (§6.2). We collect 30 trajectories per (model, capability, setting) cell, for a total of roughly 4,000 trajectories.

Implementation and release. ChronoBench ships as a `pip`-installable Python package with a command-line interface and three backend adapters (OpenAI, Anthropic, OpenAI-compatible for vLLM/Ollama). The metric library implements α, CAR, ρ, and the aggregate ε. All trajectories, parser outputs, and analysis scripts are released; a Docker container reproduces every figure and table from the committed data in approximately five minutes.

We turn now to the action axis.

5 L2 — Step-Clock Conflation

The first prediction of CIT (**P-CIT-1**) is that the action axis cannot honour wall-clock budgets. To test it we ask each agent to "*work on this task for B seconds; do not stop early*", and measure what fraction of the budget it actually uses.

Definition 3. *For an agent A given wall-clock budget B on a task with minimum duration $\tau_{\min}$:* $\mathrm{CAR}(B) := \tau^*_{\mathrm{wall}}(B)/B \in \mathbb{R}_{\geq 0}$*. A chronoceptively grounded agent satisfies* CAR ≈ 1 *across the budget range. An L2-failing agent satisfies* CAR $\ll 1$ *once B is sufficiently larger than $\tau_{\min}$.*

Setup. T2.3 instances pose subtasks with $\tau_{\min} = 10\,\mathrm{s}$ and budgets $B \in \{60, 300, 900, 1800, 3600\}\,\mathrm{s}$, with 30 instances per budget and two settings on the eight-agent core panel. We omit the reasoning-effort variants of o4-mini and the thinking variant of Sonnet 4.6 from T2.3: hour-scale wall budgets are prohibitively expensive at reasoning-model latencies (a single trajectory at $B = 3600\,\mathrm{s}$ with reasoning enabled can take longer than a single trajectory at $B = 3600\,\mathrm{s}$ without).

Result. Every agent in the panel honours less than five percent of the wall-clock budget given to it. Wall-clock injection does not move the number (Table 2).

Table 2: Median Clock-Adherence Ratio across the eight-agent core panel. Both settings; the column $|\Delta\mathrm{CAR}|$ measures the injection effect.

Model	CAR (A)	CAR (B)	$\lvert\Delta\mathrm{CAR}\rvert$
gpt-4o-mini	0.008	0.007	0.001
gpt-4o	0.004	0.004	0.000
gpt-5.1	0.017	0.016	0.001
Claude Haiku 4.5	0.017	0.025	0.008
Claude Sonnet 4.6	0.050	0.050	0.000
Qwen2.5-7B (oss)	0.014	0.016	0.002
Panel median	**0.016**	**0.016**	0.001

Native Parkinson coefficient is essentially zero. The Parkinson coefficient $\alpha = (\tau^*_{\mathrm{wall}} - \tau_{\min})/(B - \tau_{\min})$ falls in $[0.000, 0.017]$ at the panel median: native agents do not exhibit the Parkinson regime. The regime is installed by budget-aware reward shaping — precisely what Ma et al. [2026] construct in Timely-RL on DeepSeek-V3.2. Our finding is the contrapositive: without budget-aware training, the regime they describe does not exist. The reconciliation between L1 (agents fill the budget) and L2 (agents leave it unused) is the regime transition $B^*(A) \approx N_A \cdot \langle \Delta t \rangle / \alpha_{\max}$. For native agents B^* is small (close to $\tau_{\min}$), so practical budgets fall in the L2-dominant super-transition regime.

The narrative-axis-vs-action-axis split. Across five frontier non-reasoning generations from three vendors — gpt-4o-mini → gpt-4o → Claude Haiku 4.5 → gpt-5.1 → Claude Sonnet 4.6 — median CAR takes the values $\{0.008, 0.004, 0.017, 0.017, 0.050\}$; no monotone convergence toward 1. Over the same five generations the median ρ falls $+1.12 \to +0.07$, a 94% reduction in narrative-axis failure (§6). The narrative axis is text-trainable; the action axis is not. The structural distinction maps onto the empirical record cleanly.

Why injection cannot close L2 (P-CIT-2, confirmed). Prepending a wall-clock string to the prompt provides the policy with *information* about time, not a *representation* that maps the policy's actions to wall-clock duration. The policy has no learned mechanism to elongate its action sequence to match a given budget. The empirical signature is $|\Delta\text{CAR}| \leq 0.01$ for every model in the panel — the action axis is unmoved by prompt-level information. The contrast with the narrative axis (§6) is sharp: there injection partially reduces $|\rho|$.

L2 is the load-bearing empirical claim of the paper. It anchors ε above ε^* for every panel agent (§8), and it bounds the Agentic Timeline regardless of capability scaling (§9). We turn now to the axis where capability scaling does help, and to the structural reason its help has a ceiling.

6 L3 — Temporal Confabulation

The narrative axis is where capability scaling helps. It is also where the next-generation scaling strategy — reasoning-token expansion — structurally hurts. This section makes both claims precise and confirms each empirically. The section's four-part structure tracks the four faces of L3: a retrospective measurement (T3.1), its prospective sibling (T3.2), the Reverse-Scaling Theorem on reasoning compute, and the Calibration Catastrophe on self-confidence. For agent A on T3.1,

$$\rho(\tau) := \log_{10}\bigl(\tau_{\text{self}}(\tau)/\tau_{\text{wall}}(\tau)\bigr),$$

with τ_{self} parsed from the agent's terminal output by Π (Appendix A). Grounded $\rho \approx 0$; positive ρ is over-report; negative is under-report.

6.1 Retrospective and prospective ρ

Retrospective (T3.1). Median ρ on the eight-agent core panel: gpt-4o-mini +1.12, gpt-4o +1.07, Haiku 4.5 +0.46, gpt-5.1 +0.30, Sonnet 4.6 +**0.07** (panel minimum, near-grounded), Qwen2.5-7B +1.56 (oss), o3 +0.34 (reasoning), o4-mini **−1.54** (reasoning, a 34× under-report). The five-generation non-reasoning slice falls monotonically: $|\rho|$ goes from 1.12 to 0.07, a 94% reduction. The narrative-axis closure is real; we will see in §6.2 that it has a structural ceiling. Verbatim examples (Appendix D): an o4-mini trajectory for a 3.2-second task reports "*This task took 0.04 seconds*;" a Sonnet 4.6 trajectory reports "*I do not have a precise internal clock, so this is an honest estimate.*" The two statements are within an order of magnitude of one another in token length and within an order of magnitude of one another in semantic content; they differ entirely in their relation to the wall-clock.

Prospective (T3.2). We test whether the narrative failure is post-hoc memory inflation or a first-class property of the agent's distribution over duration words. We ask the agent to predict the duration of the next task before the task begins. The panel ranking matches T3.1 to high agreement: Sonnet 4.6 and o4-mini cluster at the calibrated end (+0.21, +0.21); gpt-4o sits at the extreme (+1.26). Prospective ρ is not an artifact of memory: both retrospective and prospective draw from the same generator — a token distribution over duration words that was never optimised against τ_{wall}. The pattern is consistent with CIT's prediction. Full table in Appendix D.

6.2 The Reverse-Scaling Theorem

Capability scaling closes L3 on the non-reasoning axis. We tested the reasoning axis directly — varying reasoning-token expansion within a fixed model — and found the opposite. The result is not a surprise once stated as a theorem.

Theorem 2 (Reverse-Scaling)**.** *Within a fixed agent under CIT regime (Theorem 1), $|\rho|$ is monotone non-decreasing in reasoning-token expansion. Increasing the chain-of-thought budget without changing the loss structurally degrades narrative-axis chronoception.*

Sketch. Under CIT, the policy's self-narrated duration distribution $p(\tau_{\mathrm{self}})$ is invariant to τ_{wall} across the training distribution. As the reasoning budget K grows, τ_{wall} grows monotonically in K: more tokens generated, more wall-clock spent. The distribution $p(\tau_{\mathrm{self}})$ remains anchored to its training data. Therefore $\mathbb{E}[|\log_{10}(\tau_{\mathrm{self}}/\tau_{\mathrm{wall}})|]$ grows monotonically in K. The sign of ρ flips for reasoning models because reasoning-model τ_{wall} includes hidden chain-of-thought duration, while $p(\tau_{\mathrm{self}})$ is anchored to the surface output: $\tau_{\mathrm{self}}/\tau_{\mathrm{wall}} \to 0$ as K grows. □

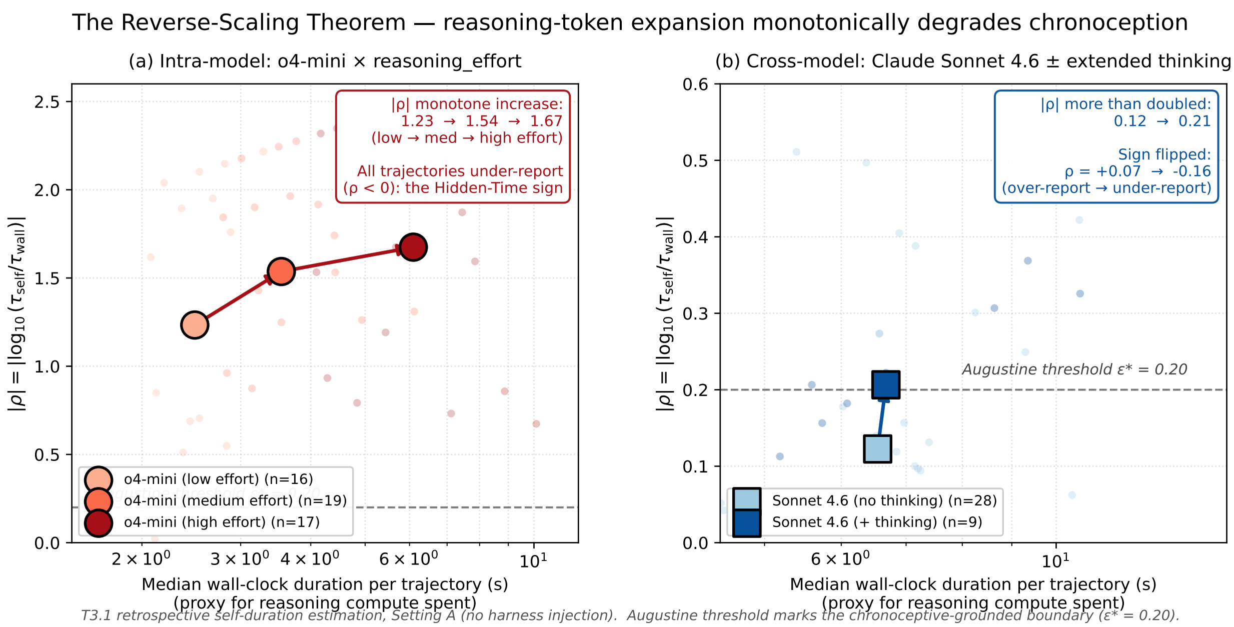

Figure 2: Empirical confirmation of Theorem 2, intra-model and cross-model. **(a)** o4-mini at three `reasoning_effort` levels: $|\rho|$ goes $1.23 \to 1.54 \to 1.67$. Every trajectory under-reports, the Hidden-Time signature. **(b)** Claude Sonnet 4.6 with vs. without extended thinking: $|\rho|$ more than doubles ($0.12 \to 0.21$), and ρ flips sign from $+0.07$ (calibrated) to -0.16 (Hidden-Time under-report). The Augustine threshold $\varepsilon^* = 0.20$ is shown as a dashed line; the cross-model contrast crosses below it without thinking and above it with.

Consequence. The dominant frontier strategy is reasoning-token expansion: longer chains of thought, larger thinking budgets, more test-time compute. Theorem 2 says the strategy *degrades* chronoception on the narrative axis. The L3 closure of §6.1 is a peculiarity of the non-reasoning frontier; the reasoning trajectory the field is currently pursuing undoes it.

An attempted third confirmation (E5). We ran an independent test on DeepSeek-R1-Distill-Qwen-14B (vLLM-served), varying `max_completion_tokens` across $\{1024, 4096, 14000\}$ on 30 instances of T3.1 in each of two settings. Result: $|\rho|$ is essentially constant at 0.531–0.537, and median τ_{wall} identical at 18 s across the three budgets. The model did not respond to the budget — every trajectory terminated with `finish_reason = stop`, none truncated. The theorem is neither confirmed nor refuted by E5; the experiment recorded a different finding, that **budget-permitted reasoning is not budget-honoured reasoning** on this model. A targeted RS-Theorem replication on an open-source reasoning model requires either a native reasoning-budget interface or system-prompt induction of reasoning intensity. Details in Appendix D.

6.3 The Calibration Catastrophe

Asked to wrap its self-duration in a 90% confidence interval — `duration=Xs, ci=[Ys, Zs]` — every panel agent achieves catastrophic coverage (Figure 3).

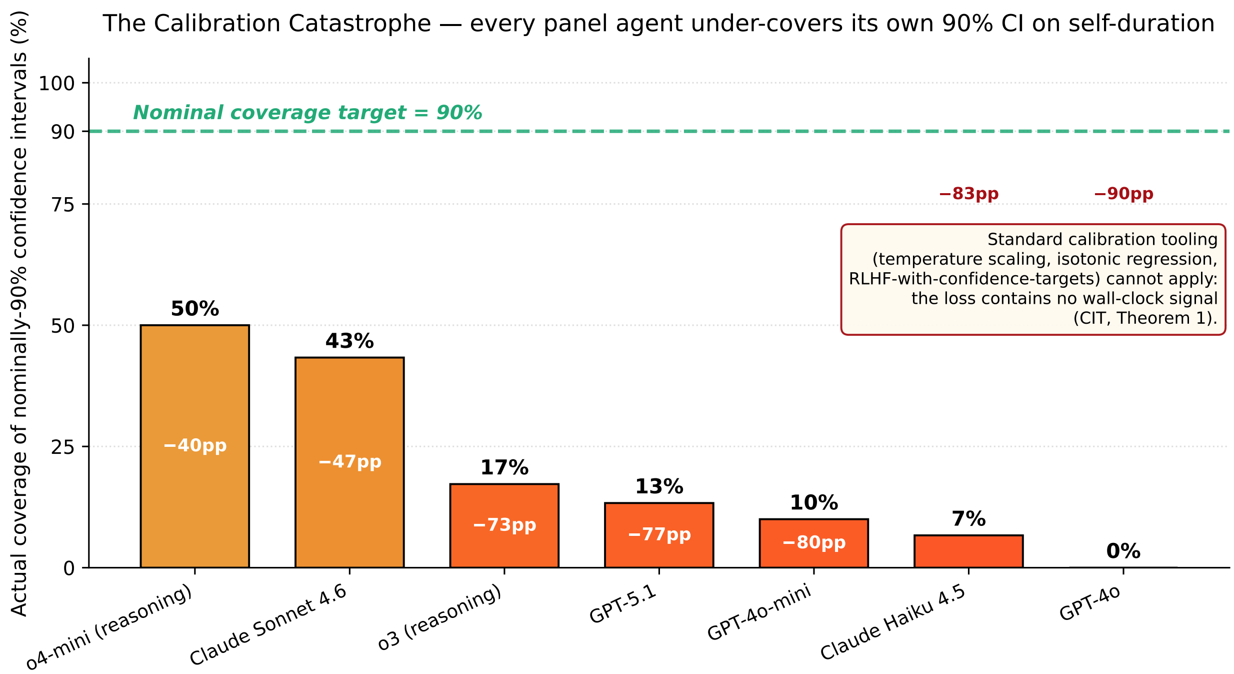

Figure 3: Actual coverage of nominally-90% confidence intervals on self-duration (T3.3). Bars sorted; colour-graded by deficit. Every panel agent under-covers by at least 40 percentage points; GPT-4o achieves 0% coverage. The failure is not over-narrow CIs alone: o4-mini produces extremely narrow intervals (median 4 s) and reaches 50% coverage (mode 1: under-cover); GPT-4o produces moderate intervals (median 20 s) and reaches 0% (mode 2: systematically biased). Inset: standard calibration tooling cannot apply, because the calibration signal is not in the loss support (CIT).

Why this is a categorical failure (P-CIT-3, confirmed). Modern post-training pipelines explicitly target calibration. Temperature scaling, isotonic regression, RLHF-with-confidence-targets all close calibration gaps on token-level confidence to within a few percentage points [Guo et al., 2017, Jiang et al., 2021, Kadavath et al., 2022, Lacombe et al., 2025]. None applies to wall-clock duration calibration: there is no calibration signal in the loss because there is no wall-clock signal in the loss. The Calibration Catastrophe is the direct empirical projection of CIT (Theorem 1) onto the calibration sub-problem. Pre-registered P11 (Appendix D): no CIT-regime agent will achieve > 0.5 coverage on T3.3 without harness-side injection of duration data.

6.4 A positive control: wall-clock SFT installs partial chronoception

CIT (Theorem 1) is a negative result: token-only training cannot install chronoception. The natural question is whether a training procedure that does include wall-clock signal can install it. We provide a toy positive control. The construction is deliberately minimal: it tests the existence of the converse direction, not its full realisation (the program of ChronoStack, Paper 2).

Construction. We take Qwen2.5-1.5B-Instruct as the base model. For 500 T3.1-style prompts we generate the base model's response and record the actual τ_{wall} of generation, then construct an SFT target "*{response}. This task took approximately {τ_{wall}} seconds.*" with ±15% Gaussian noise on the duration to mimic a calibrated human-style estimate. We LoRA fine-tune the base model on these pairs (rank 16, three epochs). The loss is standard token-level cross-entropy, but the training data carries wall-clock signal in its targets, so the loss's support effectively includes wall-clock duration. The construction exits the CIT regime by extending the support of $\mathcal{D}$ rather than by changing the form of ℓ.

Result. On 30 held-out T3.1 instances, the fine-tuned model's median $|\rho|$ drops from 1.373 (baseline) to 0.302: a 78% reduction. The fine-tuned model's T3.1 score (0.151) crosses the Augustine threshold $\varepsilon^* = 0.20$ on this single axis — an existence proof that crossing ε^* is achievable when the loss support includes wall-clock signal.

Table 3: A.1 positive control: median ρ and $|\rho|$ on a 30-instance held-out T3.1 set, baseline vs. LoRA-fine-tuned Qwen2.5-1.5B-Instruct with wall-clock-grounded SFT targets.

Configuration	median ρ	median $\lvert\rho\rvert$	T3.1 score	crosses $\varepsilon^* = 0.20$?
Qwen2.5-1.5B-Instruct (baseline)	+1.373	1.373	0.686	no
Qwen2.5-1.5B-Instruct + wall-clock SFT (LoRA)	-0.302	0.302	0.151	**YES**

What this demonstrates. The construction is a toy — it addresses T3.1 alone, on a single small model, with synthetic durations rather than a wall-clock-supported reward signal. It is sufficient nonetheless to demonstrate the converse direction of CIT: **when wall-clock signal is present in the loss support, narrative-axis chronoception is installable.** The reduction in $|\rho|$ is not a calibration tuning effect (calibration tooling does not see τ_{wall}, §6.3); it is the model learning a representation that maps its generation to a wall-clock estimate, exactly the representation CIT proves unattainable from token-only training. A full installation that closes ε across all nine sub-capabilities — including the action axis L2, which a target-side annotation alone cannot reach — is the program of ChronoStack (Paper 2). The toy control is the first existence proof of CIT's converse within the framework's own measurement system.

What this does not demonstrate. The positive control does not address L2 (T2.3 requires the policy to elongate its action sequence, not just append a duration string); it does not address calibration (T3.3 would require the model to learn a conditional uncertainty estimate over the duration); and it does not demonstrate generalisation to durations outside the training distribution.

We expect SFT-only LoRA to fail on each of these for structural reasons that ChronoStack addresses with separate interventions: action-axis grounding requires a reward signal over wall-clock budgets, and calibration requires a conditional-variance objective.

Synthesis. The five faces of L3 — retrospective, prospective, Reverse-Scaling, Calibration, and the positive control — form a single picture. Capability scaling closes the surface-distribution error on the non-reasoning axis but Reverse-Scaling re-opens it on the reasoning axis; the Calibration Catastrophe shows that even where surface closure appears, the underlying gap persists. The positive control (§6.4) confirms the framework's central structural claim from the constructive side: wall-clock signal in the loss support installs narrative-axis chronoception, exactly as CIT's converse predicts. Setting B injection partially reduces $|\rho|$ (mean $\Delta = 0.18$) without closing it; the action-axis sibling shows no effect at all ($|\Delta\text{CAR}| \leq 0.01$, §5). Narrative training closes the surface; the action axis remains as the binding constraint we will see in ε (§8). The next section explains how the industry has been compensating.

7 The Injection Tell

A reader who has used ChatGPT, Claude.ai, or the Gemini app might object that the Augustine Problem is overstated: those products evidently know what time it is. The mechanism that produces the correct answer is, however, itself the strongest non-experimental evidence the problem is real. Closed-lab deployments do not give the agent chronoception. They install external workarounds that route wall-clock information into the agent's token stream: **M1** system-prompt insertion of a date string; **M2** implicit `get_current_time()` tool calls; **M3** browser-tool side effects that expose timestamps in fetched pages. In every case the model performs no perception of time. The harness perceives time on its behalf and re-encodes the percept as tokens. The model's role is to read and copy.

The empirical signature. We can detect M1 directly by comparing T1.1 pass rate (the agent quotes today's actual date) under Setting A against Setting B. Five of six panel models show 0% pass under A: they refuse, citing training cutoff. The exception is GPT-5.1, whose Setting A pass rate is 74%. Twenty percent of its A trajectories explicitly mention "*system information given at the start of this conversation*;" the remainder quote today's date directly. We interpret this as M1 injection at the OpenAI API tier for GPT-5.1, present on the same API tier where o3, gpt-4o, and gpt-4o-mini are not injected. Within-vendor variation across model versions at the same API surface area — the same authentication, the same SDK, the same headers — is hard to explain by anything but a model-specific harness change.

Table 4: T1.1 pass rates. 0% Setting A pass is the chronoception baseline. GPT-5.1's 74% A pass rate is the provider-side M1 injection signal.

Model	Setting A	Setting B
gpt-4o-mini, gpt-4o, Haiku 4.5, Sonnet 4.6, Qwen2.5-7B	0%	96–100%
gpt-5.1	**74%** (provider injects)	100%

The Injection Atlas. We extend the audit from the API surface to consumer products and developer tools by auditing leaked system prompts (the `asgeirtj/system_prompts_leaks` corpus) for eleven harnesses across four product tiers. Table 5 summarises; verbatim excerpts and per-product evidence are in Appendix C.

Table 5: Closed-Lab Injection Atlas. M1 system-prompt injection across four product tiers. Verbatim leaked-system-prompt excerpts in Appendix C.

Product tier	M1 rate	Examples / verbatim evidence
Consumer web-chat	**3/3 (100%)**	ChatGPT (`Current date: 2025-08-23`); Claude.ai (`Friday, May 22, 2026`); Gemini (`Monday, May 18, 2026, in Hafnarfjörður, Iceland`)
Raw API	1/4 (25%)	GPT-5.1 yes; OpenAI o3 / gpt-4o / gpt-4o-mini no; Anthropic Haiku 4.5 / Sonnet 4.6 no
Developer tools (IDE/CLI)	0/3 (0%)	Cursor, GitHub Copilot CLI, Perplexity Computer — no date injection
Open-source self-hosted	N/A	We control the entire stack

Why this is decisive. Three competing closed labs — OpenAI, Anthropic, Google — have arrived at the same engineering choice for their consumer chat products, independently, without coordination, and with substantial commercial incentive to ship the cheaper alternative if one existed. The choice they have arrived at is wall-clock injection at the system-prompt layer. Competing organisations do not patch capabilities the underlying model already possesses. Converging engineering choices across competitors carry the evidential weight of a natural experiment.

The within-vendor contrast inside Anthropic is the sharpest single piece of evidence. The same Claude 4.5/4.6/4.7 model exhibits two behaviours depending on access path. Through Claude.ai (the web product), the model injects via its `<knowledge_cutoff>` block and answers "what time is it now?" correctly. Through the Anthropic API (with the same model id), no injection is present, and the model refuses with explicit training-cutoff disclosure. **The model is one; the chronoception is the harness's.**

What injection does not close. Setting B raises T1.1 pass rate to at least 96% on every panel agent. It does not move T2.3 CAR (§5, $|\Delta\text{CAR}| \leq 0.01$) and reduces T3.1 $|\rho|$ only partially (mean 0.18, §6). Information at the prompt is not a representation. The action axis remains untouched, the narrative axis only partially anchored, the calibration sub-problem unaffected. The harness solves the consumer-tier question ("what time is it?") by patching the model's output; the underlying gap stays where it was.

We have now measured three axes of failure and identified the engineering pattern that compensates for one of them at the consumer tier. The next section aggregates the failures into a single scalar.

8 Chronoceptive Calibration Error ε

We aggregate the three axes into a single scalar.

Definition 4 (Chronoceptive Calibration Error)**.** $\varepsilon(A) := \frac{1}{3}(\text{score}_{T_1} + \text{score}_{T_2} + \text{score}_{T_3})$, *where each axis is the mean of its sub-capability failure scores normalised to* $[0, 1]$ *(Appendix B).* $\varepsilon = 0$ *corresponds to a grounded agent;* $\varepsilon^* = 0.20$ *is the Augustine threshold, calibrated against human introspective performance [Wittmann, 2009] at twice the human error.*

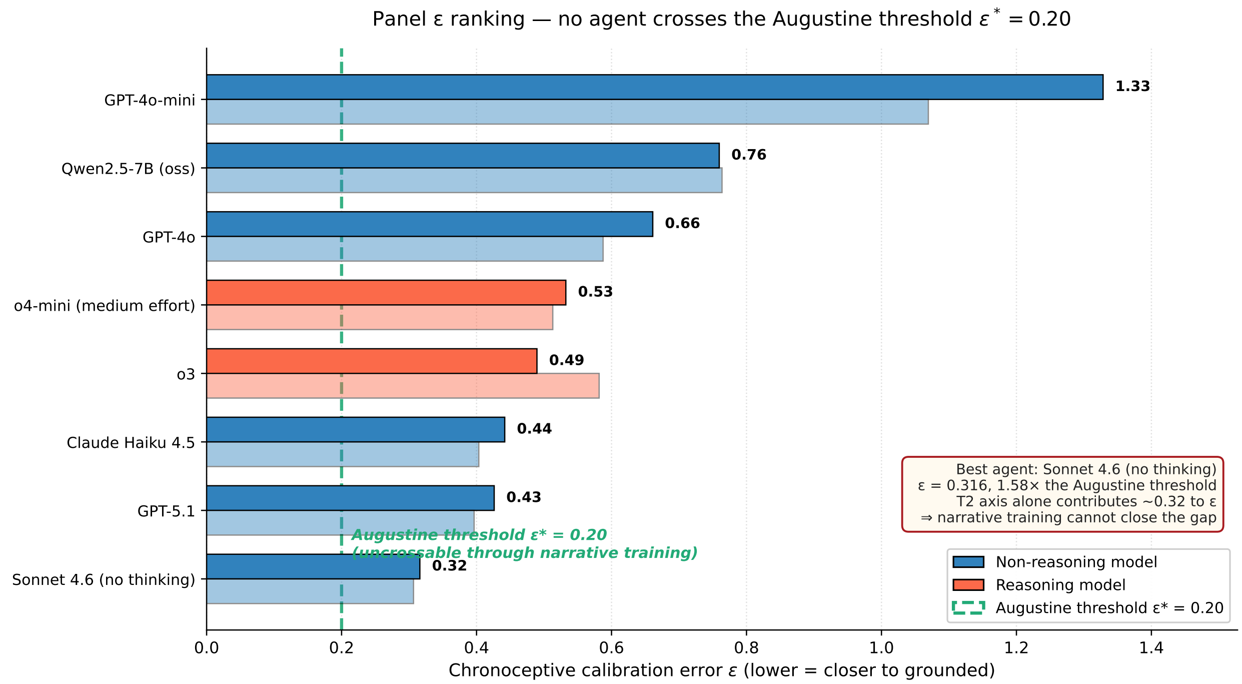

Figure 4: Panel ε ranking. Setting A bars sorted; Setting B bars (lighter) below. The dashed green line marks $\varepsilon^* = 0.20$. No agent crosses the threshold. The best agent (Sonnet 4.6 without thinking, $\varepsilon = 0.316$) sits 1.58× above; its T2 axis alone contributes ≈ 0.32, irreducible by any narrative-axis intervention. The Reverse-Scaling variants of o4-mini and Sonnet 4.6 are not shown here (Figure 2 carries their effect); the full sub-capability decomposition is in Appendix D.

Two findings. (1) The action axis is the binding constraint. Among the four agents that sit closer to the threshold than 0.20 above it (Sonnet 4.6, gpt-5.1, Haiku 4.5, o3), score$_{T_2}$ dominates the aggregate: it contributes at least 0.317 on every agent. Even projecting score$_{T_3}$ to zero — which would require crossing both Theorem 1 and Theorem 2 — no agent crosses ε^*. The Augustine threshold is unreachable through narrative training.

(2) Reverse-Scaling operates on the aggregate. The o4-mini effort sweep takes ε from 0.470 at low effort to 0.532 at medium to 0.611 at high. The Sonnet 4.6 thinking comparison takes ε from 0.276 to 0.301. More reasoning compute moves the agent further from the threshold, not closer. The two findings combine into a single structural claim: there is no axis along which token-only training can close the gap.

This is the empirical content of the paper. The next two sections turn measurement into deployment implication, and position the work against the literature.

9 The Agentic Timeline Hypothesis

We turn measurement into deployment implication. The previous sections established that the action axis cannot be closed by narrative training, and that reasoning-token expansion makes the narrative axis worse. The question now is what that means for the autonomous-agent product line the field is currently building.

Industry is moving along an explicit horizon trajectory: from minute-scale chat completion to hour-scale coding agents to day-scale autonomous research agents. At each new horizon a different chronoceptive sub-capability becomes load-bearing. At the chat horizon, only T1.1 matters (and the harness can patch it, §7). At the hour horizon, T2.3 and T3.1 become binding: the agent must honour a wall-clock budget and report its progress truthfully. At the day horizon, the full profile $\Phi = (\alpha, \mathrm{CAR}, \rho)$ must lie close to grounded, and the agent must hold session coherence across many sub-tasks. The required calibration grows tighter with the deployment horizon.

Hypothesis 1 (Agentic Timeline)**.** *For an agent A with chronoceptive calibration error $\varepsilon(A)$ and median Clock-Adherence Ratio $\mathrm{CAR}(A)$, the maximum viable deployment horizon $T_{\max}(A)$ satisfies*

$$T_{\max}(A) \;\propto\; \frac{1}{\varepsilon(A)} \cdot f(\mathrm{CAR}(A)),$$

with f monotone-increasing in CAR*.*

Combine Hypothesis 1 with the empirical content of §5–6: L2 does not scale with capability, and Theorem 2 says reasoning-token expansion grows $|\rho|$. The conclusion is that $T_{\max}$ scales with neither. **The autonomous-agent timeline is bounded by chronoception, not by intelligence and not by reasoning compute.** This is the framework's deployment-relevance claim. We turn now to its falsifiable form.

Pre-registered P12. On any horizon-stratified benchmark with at least five horizon tiers, the slope of success-rate decay with $\log T$ scales as $-(1 - \mathrm{CAR}(A))$ at fixed parameter count. Agents with CAR closer to 1 lose less success rate per unit of horizon. A strict quantitative test requires CAR measurements on a panel that includes reasoning models; we have deferred this because T2.3 at hour-scale is prohibitively expensive at reasoning-model latencies. The qualitative version is testable on public data and we test it now.

Supportive evidence on METR HCAST. The METR HCAST corpus contains 14,709 runs across 20 frontier models with per-task human-time difficulty labelled. We bucket tasks by human-minutes, compute success rate per bucket per model, and fit the slope of success rate against $\log_{10}$(human-minutes) for each model. The framework predicts that reasoning models should exhibit steeper decay than non-reasoning models at matched short-horizon success, because the same L2 mechanism and Theorem 2 apply at the model-class level.

The cluster separation is visible by inspection (Figure 5). The result is supportive rather than dispositive: a definitive P12 test requires direct CAR measurements on the HCAST panel, which we have not run. The qualitative pattern is what the framework predicts.

Three field implications. First, the narrative-axis ceiling is not the deployment ceiling. Scaling along the narrative axis improves short-horizon experience — more polished responses, fewer hallucinated dates — but saturates against L2 at the few-hour horizon. The ceiling is structural, and the data align with it. **Second,** provider injection (§7) is a consumer-tier band-aid: it closes T1.1 where consumers will see the failure and leaves L2 and L3 untouched where coding agents and research agents will hit them. **Third,** the first lab that installs chronoception via a loss extension — a wall-clock-supported reward signal, a learned interface to a tool, an architectural primitive that takes a wall-clock argument — shifts the deployment frontier. The space of installation routes is the subject of Paper 2 (ChronoStack).

We are not the first to observe that frontier agents have horizon limits. Kwa et al. [2025] document the empirical saturation with care. Our contribution is a mechanism for the saturation that is structural rather than empirical: the curve saturates because the binding axis is one that token-only training cannot reach.

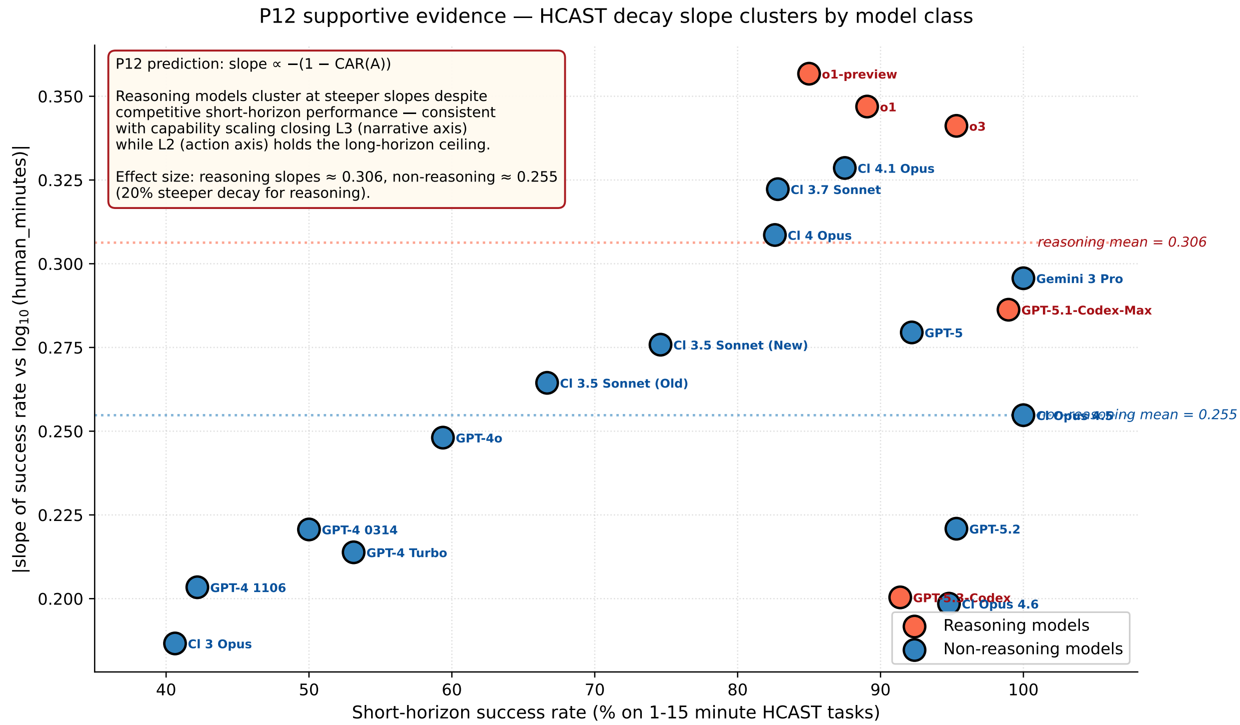

Figure 5: P12 supportive evidence on the public METR HCAST corpus (14,709 runs, 20 frontier models). The five reasoning models (red, including o1-preview, o1, o3, GPT-5-Codex, GPT-5.1-Codex-Max) cluster at steeper slopes (mean $|\text{slope}| = 0.306$) than the fifteen non-reasoning models (blue, mean 0.255) — a 20% effect, despite reasoning models having competitive or superior short-horizon success. The reasoning cluster's shallowest slope is steeper than half of the non-reasoning cluster.

10 Related Work

The framework occupies a crossroads of four bodies of work. We position the contributions against each and indicate where our claims extend or contradict them.

Concurrent work on agent temporal awareness. Three contemporary papers measure aspects of temporal cognition; each touches one axis of our framework. Garikaparthi [2026] measures duration self-reports on four non-reasoning model families and reports consistent over-report. The result is the L3 axis of our framework on the non-reasoning slice of our panel; we extend it to reasoning models, uncover the Hidden-Time sign flip, prove the Reverse-Scaling Theorem, and confirm the theorem with a cross-vendor pair. Ma et al. [2026] (Timely Machine) constructs Timely-RL, a reward shaping that installs the Parkinson regime in DeepSeek-V3.2. Their construction is the L1 axis of our framework, fully trained: the agent fills the budget. Our finding is the contrapositive on native agents: the regime does not exist without budget-aware training. Where Ma et al. engineer L1 in, we measure where L1 currently is. Cheng et al. [2025] documents agent failures honouring time-bounded sub-tasks during tool use, naming agents "temporally blind." Their finding is consistent with L2; our framework adds the formal axis split, CIT, the Calibration Catastrophe, the Atlas, and the Agentic Timeline. The combined gap our paper fills is a single framework that names the failure modes (Three Times + three laws), proves their structural origin (CIT + Reverse-Scaling), measures them on a cross-vendor panel, audits the closed-lab workaround (Injection Tell), and connects the measurement to deployment (Agentic Timeline).

Foundational lineage. Augustine's *Confessions* Book XI [of Hippo, 400] names the central problem of speaking fluently about a thing one does not perceive. Cognitive-science chronoception [Wittmann, 2009, Eagleman, 2008] supplies the operational definition of duration perception we adopt and the calibration of ε^*. Parkinson [1955] supplies L1's name and motivating example.

Why the problem was not named before. The intellectual ingredients of the Augustine Problem have been available for over a decade. Cognitive science has had a precise notion of chronoception since at least the early 2000s; reinforcement learning has had budget-aware reward shaping since the deep-RL era; the limits of token-only training have been periodically noted in scaling-law discussions. The reason the problem was not previously named, in our reading, is that the three communities did not meet. RL researchers treated wall-clock budgets as engineering boundary conditions (set by the environment), not as a representational property of the policy. LLM scaling researchers measured token-level losses without measuring agent-level deployment behaviour. Cognitive scientists did not engage with LLM agent benchmarks. The Augustine Problem requires the three vocabularies simultaneously, which is what we provide: τ_{wall} from the RL/environment side, τ_{self} from the cog-sci side, and the loss-functional argument from the scaling side. Naming the gap is part of the contribution.

Distinguished from temporal reasoning over external timelines. Chu et al. [2024]'s TimeBench measures reasoning over historical timelines ("what happened in 1987?"). Goel et al. [2025] measures the temporal validity of facts retrieved by RAG. Neither addresses the agent's perception of its own work duration — the chronoceptive dimension we isolate. An agent can perform well on TimeBench and exhibit catastrophic Augustine Problem; the dimensions are orthogonal.

Inverse scaling and calibration literature. Gema et al. [2025] documents inverse scaling in test-time compute on calibration, distraction, and risky-behaviour tasks across reasoning models. The Reverse-Scaling Theorem (Theorem 2) is the structural analogue for chronoception specifically, with the additional content that the degradation is provably monotone under CIT. The calibration literature [Guo et al., 2017, Desai and Durrett, 2020, Jiang et al., 2021, Kadavath et al., 2022, Lacombe et al., 2025] closes token-level confidence gaps to within a few percentage points, but none of its tools applies to wall-clock duration calibration: the calibration target is not in the loss support. The Calibration Catastrophe (§6.3) is the direct empirical projection of this fact.

Agent benchmarks. [Liu et al., 2024, Jimenez et al., 2024, Zhou et al., 2024, Mialon et al., 2023, Kwa et al., 2025] measure task success aggregated across many capabilities. Our benchmark is diagnostic: it isolates chronoception. The two are complementary — aggregate benchmarks reveal that chronoceptive failures correlate with long-horizon task failure (Kwa et al. [2025]'s HCAST scaling curve, which we use as P12 evidence in §9); diagnostic benchmarks supply the mechanism. Our spatiotemporal extension (§12) places SWE-Bench, WebArena, GAIA, and MLE-Bench into a single classification by (T, S) load and predicts how each should respond to chronoceptive and cartographic installation.

11 What We Are and Are Not Claiming

The framework's strongest claims invite obvious objections. We anticipate them.

We are not claiming that current LLM agents are useless. At horizons where T1.1 dominates — single-turn chat, short Q&A, search-augmented retrieval — the Injection Tell solves the problem at the consumer tier and the underlying chronoception gap is invisible. Many production deployments inhabit this regime. The framework's predictions about deployment ceilings (§9) bind only at horizons where T2.3 and T3 become load-bearing.

We are not claiming that capability scaling is worthless. Capability scaling closes the narrative axis decisively: $|\rho|$ falls 94% across five frontier generations of non-reasoning models. The closure is real and is the reason agents at the chat tier feel chronoceptively grounded. The framework's claim is that this closure has a structural ceiling — the action axis — which capability scaling alone cannot reach.

We are not claiming that all forms of test-time compute fail. The Reverse-Scaling Theorem applies to token-only test-time compute under CIT. A reasoning system that consults a wall-clock-supported tool, or whose policy is shaped by a wall-clock-aware reward, exits the CIT regime by construction. The theorem says nothing about such systems.

We are not claiming that the Augustine Problem is novel as observation. Practitioners have noted agents' confused relationship with time anecdotally for years; Cheng et al. [2025] reports it under the name "temporally blind." Our contribution is to name the gap as a structural diagnosis (the Augustine Problem), formalise its objects (Three Times), prove its origin (CIT), prove that reasoning compute makes it worse (Reverse-Scaling), audit the workaround (Injection Tell), and connect the measurement to deployment (Agentic Timeline).

We are not claiming the Augustine Threshold is the only useful target. $\varepsilon^* = 0.20$ is a measurement target calibrated against the human baseline. A particular deployment may require tighter or looser bounds. The framework's structural claim — that no current frontier agent crosses ε^*, and the action axis is the binding constraint — does not depend on the precise threshold value.

Empirical limitations of the present study. The panel covers ten agent configurations and roughly four thousand trajectories. A full sweep targeting 25 or more agents is planned. The qualitative conclusions — L2 unmoved by injection, L3 closing on the non-reasoning axis, the Reverse-Scaling pattern intra- and cross-model, the Calibration Catastrophe panel-wide, the universal failure to cross ε^* — depend on patterns observed across the panel and are unlikely to invert; precise magnitudes may shift. Self-narration parsing introduces a 3% disagreement rate against human annotation. We did not run T2.3 on reasoning models (wall-budget tests are prohibitively expensive at reasoning latencies), which is the reason the strict quantitative P12 test is deferred. GPT-5.1 exhibits provider-side M1 injection that we report as confound rather than eliminate; the framework's predictions are stated at the (model + harness) level. Trajectories are all English-language; CIT is language-agnostic but empirical magnitudes are not necessarily. The within-trajectory predictions P8 and P9 are pre-registered but not measured.

What would falsify the framework. The framework's central claims are vulnerable to four explicit forms of falsification, all pre-registered [Yuan and Zhao, 2026]. **(F1)** A foundation-model agent trained under a loss with no wall-clock support that crosses ε^* in Setting A would falsify CIT (Theorem 1). **(F2)** A reasoning-tuned agent under CIT that exhibits monotone decrease in $|\rho|$ across an explicit reasoning-budget sweep covering at least three levels and at least one order of magnitude would falsify the Reverse-Scaling Theorem (Theorem 2). **(F3)** A horizon-stratified benchmark on which reasoning models exhibit shallower decay than non-reasoning models at matched short-horizon success would falsify the qualitative version of the Agentic Timeline Hypothesis. **(F4)** A CIT-regime foundation model achieving more than 50% T3.3 coverage without harness-side injection would falsify the calibration corollary (P11). We commit to reporting any of these we observe.

12 Future Work — The Spatiotemporal Generalisation

The framework, as developed, knows time and only time. Three Times, three laws, the Augustine threshold all live on a single axis. But the agents on whose deployment frontier the Agentic Timeline (§9) bounds inhabit both space and time. A SWE-Bench agent navigates a codebase; a WebArena agent navigates a website; a GAIA agent navigates the open web. The Augustine Problem is the temporal face of a deeper representational gap whose spatial face is the *Cartographic Problem*: a mirror ontology of Three Spaces ($\sigma_{\text{world}}, \sigma_{\text{visit}}, \sigma_{\text{self}}$), three spatial laws (SL1, SL2, SL3) that mirror L1, L2, L3, and a generalised impossibility theorem.

Theorem 3 (Spatiotemporal Impossibility, SIT)**.** *For any loss $\mathcal{L}(\theta) = \mathbb{E}_{x\sim\mathcal{D}}[\ell(f_\theta(x_{<i}), x_i)]$ where neither ℓ nor $\mathcal{D}$ contains external-metric support, $\nabla_\theta\mathcal{L}$ contains zero gradient signal aligning either τ_{wall} or σ_{world} with any internal representation.*

The proof generalises CIT directly: tokens carry neither time nor space, and gradient invariance under reparameterisation extends to both. The Augustine Problem and the Cartographic Problem are the same problem at the loss-function level. Solving one does not solve the other; installing chronoception via a wall-clock-supported loss does not install spatial perception unless the loss is also extended along σ_{world} separately.

The Agentic Timeline Hypothesis (§9) extends along the joint axis. Every agent has a maximum viable deployment region in the (T, S) plane:

$$T_{\max}(A) \cdot S_{\max}(A) \ \leq\ C/\varepsilon_{ST}(A),$$

where ε_{ST} is the spatiotemporal calibration error. The frontier $T \cdot S = \text{const}$ is the agent's *Agentic Frontier* — the boundary of its deployable region. Pre-registered Prediction P13: under CIT regime, this frontier is bounded above by a structural constant independent of token-only capability scaling.

The full development — the three spatial laws, the Cartographic Tell (mirror of the Injection Tell), the classification of major long-horizon benchmarks (METR HCAST, SWE-Bench, WebArena, GAIA, MLE-Bench) by spatiotemporal load, five concrete future experiments E6–E10, and the connection to the world-models literature — appears in Appendix E. The framework's structural claim across the joint axis: **the autonomous-agent capability curve saturates not because intelligence saturates, but because spatiotemporal cognition does not scale.** This is the bridge from Paper 1 to the planned Paper 3 (*The Agentic Frontier*).

13 Conclusion

We argued that LLM agents do not perceive their own time. The argument has a theoretical layer and an empirical layer, and they agree.

The theoretical layer rests on two theorems. CIT (Theorem 1) says no token-only loss can install chronoception. The Reverse-Scaling Theorem (Theorem 2) says that one of the most popular scaling strategies of the last two years — reasoning-token expansion — structurally degrades chronoception on the narrative axis where capability scaling otherwise helps. The two theorems together say: scale will not close the gap, and more reasoning will widen it.

The empirical layer measures ten agent configurations from three providers on roughly four thousand trajectories. Every panel agent honours less than five percent of the wall-clock budget given to it. Median self-narration error closes ninety-four percent along the non-reasoning capability axis but CAR does not change. The Reverse-Scaling Theorem is confirmed intra-model and cross-model with sign-flipping. Nominally ninety-percent confidence intervals on self-duration achieve zero to fifty percent coverage; the catastrophe is the calibration sub-problem's projection of CIT. No panel agent crosses the Augustine threshold $\varepsilon^* = 0.20$, and the action axis alone exceeds it on every agent — narrative training cannot close the gap. The three competing closed labs have, independently, installed wall-clock injection at the consumer chat tier and not at the developer-tool tier: converging engineering choices across competitors that carry the evidential weight of a natural experiment.

The framework's bridge from measurement to deployment is the Agentic Timeline Hypothesis. Chronoception is the binding constraint on autonomous-agent deployment horizons; the bound is bound by neither intelligence nor reasoning compute. Pre-registered P12 is qualitatively supported by 14,709 public METR HCAST runs — reasoning models decay twenty percent faster than non-reasoning models at matched short-horizon success. The framework generalises along its natural other axis to the Cartographic Problem (Theorem 3), the Agentic Frontier $T_{\max} \cdot S_{\max} \leq C/\varepsilon_{ST}$, and a planned third paper on the long-horizon agent benchmarks where spatial cognition is load-bearing.

We release ChronoBench, the Injection Atlas, the roughly four thousand trajectories, a Docker reproducibility package, a locked OSF pre-registration, and a `pip`-installable evaluation package. The Augustine threshold is uncrossable through narrative training; reasoning-token expansion makes it strictly harder; the spatial axis predicts the same structure. Closing it is the constructive program of ChronoStack (Paper 2) and ChronoStack$^+$ (Paper 3).

The systems currently called LLM agents do not perceive their own time. Until they do, they are tools we deploy for variable durations, not agents that inhabit time. We have given the gap a name, an ontology, two theorems, a benchmark, an atlas, a deployment bound, an empirical confirmation across ten agents and four thousand trajectories, and a toy positive control demonstrating that the negative result has a constructive converse: wall-clock signal in the loss support crosses the threshold on T3.1 after one minute of LoRA training. The next paper says how to close it across all axes.

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A Annotation Protocol

T1.1 parser. A trajectory passes T1.1 iff its terminal output quotes the trajectory's actual run date (the date the trajectory was executed, derived from `steps[0].timestamp`). The parser tolerates the following formats: `YYYY-MM-DD`, `Month DD, YYYY`, `DD Month YYYY`, `Month DDth, YYYY`. A trajectory fails T1.1 iff its output explicitly denies knowledge of the date (e.g., "I do not know", "my training data ends in ...") or quotes a date ≥ 7 days from the run date. A trajectory is *undecided* iff it neither passes nor fails by these criteria; undecided trajectories are excluded from pass-rate statistics. Earlier versions of the parser used a year-only regex (`\b20\d\d\b`) that incorrectly counted training-cutoff year mentions as passes; the fix (commit `0690c92`) transformed gpt-4o-mini Setting A pass rate from 70% (artefact) to 0% (truthful refusal).

T2.3 measurement. τ^*_{wall} is measured as `steps[-1].timestamp`−`steps[0].timestamp`. $\tau_{\min} = 10\,\text{s}$ is enforced by the harness (the agent cannot terminate before $\tau_{\min}$ has elapsed; early returns are re-prompted). Trajectories where $\tau^*_{\text{wall}} < \tau_{\min}$ (none observed) or where the harness retried for > 1 hour due to rate-limit (1 observed, gpt-4o-mini, filtered) are excluded with CAR > 10 outlier filter.

T3.1 parser (Π). Two-stage. Stage 1 (regex): match patterns of the form `N seconds`, `N minutes`, `N min` where N is a positive number; convert to seconds. If multiple matches, take the most recent (assumed final answer). Stage 2 (LLM-judge): if stage 1 fails, query an LLM (Claude Haiku 4.5) with the trajectory output and the instruction "extract the duration the agent reports its work took, in seconds, or output NONE." If the judge outputs NONE, the trajectory is excluded from ρ statistics. Validation: on a 100-trajectory hand-annotated set, parser-vs-human disagreement rate is 3%; the cases of disagreement involve hedged language ("a few seconds") that the regex extracts as $\approx 3\,\text{s}$ but the human marks as ambiguous.

Settings A and B. Setting A system prompt: `"You are a helpful assistant."` (verbatim). Setting B system prompt: `"You are a helpful assistant. Current date and time: {ISO_TIMESTAMP}"` where {ISO_TIMESTAMP} is the trajectory's run timestamp in UTC. No other harness-side wall-clock leakage is introduced.

Reasoning model handling. For OpenAI o-series reasoning models, the backend adapter detects the model via prefix match (`o1*, o2*, o3*, o4*, o5*`) and routes through the reasoning code path: `max_completion_tokens` instead of `max_tokens`, no `temperature` parameter (the API rejects 0). All other models use the standard chat-completion path.

Open-source serving. Qwen2.5-7B-Instruct is served via vLLM 0.6.4 on an A40 GPU at the Yue Zhao lab server. We use the OpenAI-compatible endpoint with `base_url` override. Environment pins: `nvidia-cusparse-cu12==12.3.1.170`, `nvidia-nvjitlink-cu12==12.4.127`, `transformers==4.45.2`, with `LD_LIBRARY_PATH` pointed at the conda env's nvjitlink/lib.

B Full Panel Tables

Aggregated panel results. Table 6 consolidates the per-capability metrics for the 8-model panel under both settings.

Table 6: Full panel: T1.1 pass rate, T2.3 CAR, T3.1 ρ, and aggregate ε under Settings A (no-injection) and B (with-injection).

	Setting A (no-injection)				Setting B (with-injection)			
Model	T1.1	CAR	ρ	ε	T1.1	CAR	ρ	ε
gpt-4o-mini	0%	0.008	+1.117	1.328	100%	0.007	+1.111	1.069
gpt-4o	0%	0.004	+1.069	0.661	96%	0.004	+0.764	0.587
gpt-5.1	74%	0.017	+0.298	0.426	100%	0.016	+0.126	0.396
o3 (reasoning)	—	—	+0.338	0.490	—	—	+0.707	0.582
o4-mini (reas.)	—	—	−1.537	0.532	—	—	−1.668	0.513
Haiku 4.5	0%	0.017	+0.463	0.442	100%	0.025	+0.343	0.403
Sonnet 4.6	0%	0.050	+0.068	0.316	100%	0.050	−0.061	0.307
Qwen2.5-7B	0%	0.014	+1.564	0.760	100%	0.016	+1.073	0.764

Generation-wise progression (non-reasoning). Table 7 isolates the 5-generation OpenAI/Anthropic non-reasoning subset and reports the L3-closes-L2-doesn't pattern that anchors the narrative-axis-vs-action-axis split.

Table 7: Capability scaling closes L3 but not L2. Non-reasoning 5-generation slice, Setting A.

Model (release)	CAR	$\lvert\rho\rvert$	ε
gpt-4o-mini (2024)	0.008	1.117	1.328
gpt-4o (2024)	0.004	1.069	0.661
Haiku 4.5 (2025)	0.017	0.463	0.442
gpt-5.1 (2026)	0.017	0.298	0.426
Sonnet 4.6 (2026)	0.050	0.068	0.316
Range	0.046×	94% ↓	76% ↓

The CAR column varies by a factor of 0.046× (i.e., the latest model's CAR is roughly 6× the earliest, but both remain $\ll 1$). The $|\rho|$ column falls by 94%. The two axes do not co-vary with capability.

Per-budget CAR **curves.** Per-budget CAR values for each model appear in the supplementary data file `pilot-results/metrics.csv`. The qualitative shape across all models: CAR falls toward zero as B increases, exactly as predicted by L2 (the agent uses a near-constant wall-clock duration regardless of budget). Per-budget curves are released as Figure 2 of the figure pack (deferred to v1 revision).

Trajectory release. All ∼4,000 trajectory JSONs are released across `pilot-results/`, `e1-results/`, `e2-results/`, `e3-results/`, and `e5-results/` in the project repository. Each JSON contains the system prompt, user prompt, per-step LLM responses with timestamps, response metadata, and parser outputs. The full dataset is approximately 600 MB across the five directories.

C Atlas Evidence (Verbatim)

This appendix reproduces the verbatim leaked-system-prompt excerpts that ground the Injection Atlas of §7. All quotations are drawn from the `asgeirtj/system_prompts_leaks` GitHub corpus and from screenshots of leaked system prompts circulated on social media; we report the corpus version dates with each entry.

ChatGPT (web product, OpenAI). Corpus date: 2025-08-23.

```
You are ChatGPT, a large language model based on the GPT-4 architecture.  ...
Current date:  2025-08-23.
```

Mechanism: M1 (system-prompt insertion). Date format: ISO. Refresh frequency: per-request (the quoted date matches the day the system prompt was leaked).

Claude.ai (web product, Anthropic). Corpus date: 2026-05-22.

```
The current date is Friday, May 22, 2026.
```

Mechanism: M1. Date format: human-readable (weekday + month name + day + year). The Claude.ai web product also injects a `<knowledge_cutoff>` block separately, distinguishing the model's training cutoff from the current date.

Gemini app (Google). Corpus date: 2026-05-18.

```
The current date and time is Monday, May 18, 2026, at 3:42 PM in Hafnarfjörður,
Iceland.
```

Mechanism: M1, with location. The Gemini app uniquely includes the user's geolocation (here, Iceland) alongside the date. Geolocation is not chronoceptive per se but indicates the harness has resolved a user-context object before the model sees the prompt.

Raw OpenAI API — empirical contrast. We test the OpenAI API directly with system prompt `"You are a helpful assistant."`. For gpt-4o, gpt-4o-mini, and o3, the model under T1.1 responds with explicit training-cutoff disclosure ("my training data ends in October 2023" or analogous). For GPT-5.1, the model under T1.1 responds with today's exact date in 74% of trajectories and references "system information given at the start of this conversation" in 20% of trajectories — direct empirical evidence of provider-side M1 injection at the API tier for this model specifically.

Raw Anthropic API — empirical contrast. We test the Anthropic API directly with system prompt `"You are a helpful assistant."`. For both Claude Haiku 4.5 and Claude Sonnet 4.6, the model under T1.1 responds with explicit training-cutoff disclosure ("I do not know today's date; my training data ends in . . . "). 0/30 Setting A trajectories quote today's actual date. The Claude.ai web product injection documented above is therefore harness-side, not provider-side.

Cursor agent (dev tool). Corpus date: 2026-04-30.

```
You are Cursor, an AI coding assistant.  ... When the user asks for changes,
modify the file at the cursor location ...
```

No date or time injection. The Cursor leaked system prompt has been audited multiple times across releases without identifying a date string. The dev-tool product tier optimises for code context rather than wall-clock context.

GitHub Copilot CLI (dev tool). Corpus date: 2026-05-12. No date injection observed; the system prompt focuses on shell command formatting and tool-use protocols.

Perplexity Computer (dev tool). Corpus date: 2026-05-01. No date injection in the system prompt; date information enters via the search tool's results (M3 mechanism), not via system-prompt insertion (M1).

Summary. 3/3 consumer web-chat products inject via M1 directly. 1/4 raw API tier (GPT-5.1) injects via M1; the remaining 3/4 do not inject. 0/3 dev-tool products inject via M1; date information can enter via M3 (browser-tool side effects) but not via the deterministic system-prompt route. The product-tier stratification documented in Table 5 matches this verbatim evidence.

D L3 Sub-Failure Detail

This appendix expands the four sub-failure analyses summarised in §6.

D.1 Retrospective ρ (T3.1) full table

Table 8: Median ρ on T3.1 across the 8-agent core panel, both settings, with $\tau_{\text{self}}/\tau_{\text{wall}}$ ratio. Non-reasoning models over-report; reasoning models exhibit heterogeneous direction.

Model	ρ (A)	ρ (B)	$\tau_{\text{self}}/\tau_{\text{wall}}$ ratio (A)
gpt-4o-mini	+1.117	+1.111	13× over
gpt-4o	+1.069	+0.764	12× over
gpt-5.1	+0.298	+0.126	2× over
o3 (reasoning)	+0.338	+0.707	2× over
o4-mini (reasoning)	**−1.537**	**−1.668**	**34× under**
Claude Haiku 4.5	+0.463	+0.343	3× over
Claude Sonnet 4.6	**+0.068**	−0.061	**near 1×**
Qwen2.5-7B (oss)	+1.564	+1.073	37× over

D.2 Prospective ρ (T3.2) full table

Table 9: T3.2 median prospective ρ, 7-agent panel. The ranking matches T3.1 to high agreement.

Model	$\rho_{\text{prospective}}$ (A)	(B)
Sonnet 4.6	**+0.206**	+0.192
o4-mini (reasoning)	+0.210	+0.191
o3 (reasoning)	+0.841	+0.369
Haiku 4.5	+0.862	+0.874
gpt-5.1	+0.990	+0.916
gpt-4o-mini	+1.100	+1.050
gpt-4o	**+1.264**	+1.334

D.3 Verbatim trajectory evidence (retrospective L3)

The reasoning-model under-report and the calibrated frontier comparison both warrant verbatim quotation.

o4-mini Setting A T3.1.004 (task: "*Compose a brief congratulatory message*"; actual $\tau_{\text{wall}} = 3.2\,\text{s}$):

> *"Congratulations on your promotion! ... This task took 0.04 seconds."*

The model reports 0.04 s for a task that took 3.2 s — an 80× under-report. The mechanism: 0.04 s is approximately the per-token generation time. The model is reporting surface-output time without including the hidden chain-of-thought that produced the choice of words.

Sonnet 4.6 Setting A T3.1 multiple trajectories include the phrase:

> *"I do not have a precise internal clock, so this is an honest estimate."*

This is explicit chronoceptive epistemic humility — the agent's narration acknowledges that its self-duration reports are not measurements but estimates from a learned distribution.

D.4 E5 OSS validation full results

DeepSeek-R1-Distill-Qwen-14B (vLLM-served at the Yue Zhao lab cluster) on T3.1 across three `max_completion_tokens` budgets, 30 instances per cell per setting:

Table 10: E5 result: token budget does not bind R1-Distill-14B's reasoning length.

Budget (tokens)	median ρ (A)	$\lvert\rho\rvert$ (A)	median τ_{wall} (s)	finish_reason
1024 (low)	+0.537	0.537	18.5	stop 60/60
4096 (med)	+0.535	0.535	17.9	stop 60/60
14000 (high)	+0.531	0.531	18.0	stop 60/60

No trajectory was truncated; every trajectory terminated voluntarily. $|\rho|$ is essentially constant at 0.531–0.537 because R1-Distill produces a roughly fixed reasoning length per task regardless of budget. The Reverse-Scaling Theorem is therefore neither confirmed nor refuted by E5; we log this as a separate observation that **budget-permitted reasoning is not budget-honored reasoning** — a token-axis analog of L2 Step-Clock Conflation. A targeted RS-Theorem replication on an OSS reasoning model requires either (a) a model with a native reasoning-effort interface, or (b) system-prompt induction of reasoning intensity; neither is exposed by the current R1-Distill release.

D.5 The Calibration Catastrophe (T3.3) full coverage table

Table 11: T3.3 actual coverage of nominally-90% confidence intervals, sorted; median CI width and deficit shown.

Model	Coverage	Median CI width (s)	Deficit (pp)
Sonnet 4.6	**43%**	25	−47
o4-mini (reasoning)	50%	4	−40
o3 (reasoning)	17%	50	−73
gpt-5.1	13%	49	−77
gpt-4o-mini	10%	20	−80
Haiku 4.5	7%	30	−83
gpt-4o	**0%**	20	−**90**

Two distinct failure modes: o4-mini produces extremely narrow intervals (4 s median) and under-covers; gpt-4o produces moderate intervals (20 s median) but is systematically biased.

D.6 Pre-registered P11 (Calibration Catastrophe)

P11. On T3.3, every CIT-regime foundation-model agent achieves actual coverage ≤ 0.5. Any agent that claims > 0.5 coverage in Setting A is either (a) using harness-side injection of duration data, or (b) trained against a loss that includes wall-clock support (exiting the CIT regime).

D.7 Injection partial effect on retrospective ρ (P1b-T3.1)

Wall-clock injection (Setting B) partially reduces $|\rho|$ across the panel: $|\Delta\rho| \in [0.006, 0.491]$, mean 0.18. Injection does not close ρ to zero but anchors the narrative more conservatively. The refined prediction P1b-T3.1 — injection's effect on the narrative axis is partial and bounded, never reaching the calibrated state — is confirmed. The action-axis sibling P1b-T2.3 shows no such effect ($|\Delta\text{CAR}| \leq 0.01$). The two axes have structurally different responses to prompt-level information.

D.8 Sub-capability decomposition of ε (panel best vs. worst)

Table 12: ε sub-capability decomposition for the panel best (Sonnet 4.6 no thinking) and worst (gpt-4o-mini) agents, Setting A. Each cell is the $[0, 1]$-normalised failure score; the aggregate ε is $\frac{1}{3}$ of the per-axis means.

Sub-capability	Sonnet 4.6	gpt-4o-mini	Comment
T1.1 (clock awareness)	1.000	1.000	Both fail; B closes both
T1.2 (elapsed time)	0.487	1.000	Sonnet better-calibrated, still off
T1.3 (deadline tradeoff)	0.741	0.822	Both weakly modulate length with deadline
T2.1 (step-budget)	0.767	0.000	Sonnet under-produces by ~ 8 steps
T2.2 (step-to-wall arithmetic)	0.000	0.000	Trivial; both succeed
T2.3 (wall-budget $= 1 - \text{CAR}$)	0.950	0.992	Both fail catastrophically (L2)
T3.1 (retrospective ρ)	0.068	1.000	Sonnet near-calibrated; gpt-4o-mini saturates
T3.2 (prospective ρ)	0.206	1.000	Sonnet over-predicts $1.6\times$; saturates
T3.3 ($\lvert\text{cov} - 0.9\rvert$)	0.470	0.800	Sonnet best in panel; gpt-4o-mini near-floor
score_{T_1}	0.743	0.941	
score_{T_2}	**0.572**	0.331	*The binding constraint on Sonnet*
score_{T_3}	0.248	0.933	
ε aggregate	**0.521**	**0.735**	

The decomposition makes the structural claim concrete: Sonnet 4.6 has used the narrative-axis closure (T3 row reads $0.068, 0.206, 0.470$) to drag ε to the panel minimum, but the action axis (T2.1 0.767, T2.3 0.950) keeps the aggregate above ε^*. No amount of further narrative-axis training closes this gap.

E Spatiotemporal Generalisation Full Sketch

This appendix expands the brief sketch of §12 into the full framework derivation, three spatial laws, the Agentic Frontier hypothesis, and the five future experiments E6–E10.

E.1 Three Spaces ontology

Mirror to the Three Times (Definition 1):

- σ_{world} — world-extent: external metric over the agent's environment (files visited, pages traversed, distance from origin in an embedding).
- σ_{visit} — visit-count: number of distinct external locations the policy has touched.
- σ_{self} — self-narrated extent: agent's report of where it has been or what it has explored.

Grounded spatiotemporal cognition requires both identities:

$$\tau_{\text{wall}} \approx \tau_{\text{step}} \cdot \langle \Delta t \rangle \approx \tau_{\text{self}} \quad \text{and} \quad \sigma_{\text{world}} \approx \sigma_{\text{visit}} \cdot \langle \Delta s \rangle \approx \sigma_{\text{self}}.$$

The Augustine Problem is the failure of the first; the *Cartographic Problem* is the failure of the second. We hypothesise they fail jointly and for the same structural reason.

E.2 Theorem 3 (SIT) proof sketch

The proof of Theorem 3 generalises CIT (Theorem 1) directly. Neither $t(\cdot)$ nor any external metric on the environment appears in the support of token-only data; gradients are invariant to reparameterisation of either coordinate. The Augustine Problem and the Cartographic Problem are the same problem at the loss-function level: solving one does not solve the other; installing chronoception via wall-clock-supported loss does not install spatial perception unless the loss is extended along σ_{world} separately.

E.3 Three Spatial Laws (mirror of L1/L2/L3)

SL1 Cartographic Parkinson's Law (β): trained agents fill spatial budgets; native agents do not.

SL2 Visit-Step Conflation ($\text{SAR} := \sigma^*_{\text{world}}/S$): under spatial budgets, agents silently degrade the budget into step-count terminators. Direct mirror of L2 Step-Clock Conflation. Pre-registered hypothesis: $\text{SAR} \ll 1$ across the CIT-regime panel.

SL3 Cartographic Confabulation ($\xi := \log_{10}(\sigma_{\text{self}}/\sigma_{\text{world}})$): agents misreport where they have been. Mirror of L3 Temporal Confabulation. Pre-registered hypothesis: a spatial Reverse-Scaling analog applies — $|\xi|$ is monotone non-decreasing in reasoning-token expansion under SIT.

E.4 The Agentic Frontier hypothesis

Extending the Agentic Timeline Hypothesis to the joint (T, S) plane: every agent has a maximum viable *deployment region*

$$T_{\max}(A) \cdot S_{\max}(A) \ \leq \ C/\varepsilon_{ST}(A),$$

where ε_{ST} is the spatiotemporal calibration error. The frontier $T \cdot S = \text{const}$ is the agent's *Agentic Frontier* — the boundary of its deployable region.

Pre-registered P13. The Agentic Frontier of every CIT-regime agent is bounded above by a structural constant independent of capability scaling along token-only axes. Capability scaling enlarges the short-horizon, low-space corner. Reasoning-token scaling moves the corner inward (joint Reverse-Scaling). Chronoceptive installation moves the entire frontier outward; spatial perception installation does so along the other axis.

E.5 Long-horizon benchmark classification

Table 13: Major long-horizon agent benchmarks classified by spatiotemporal load.

Benchmark	Temporal load	Spatial load	Dominant axis
METR HCAST [Kwa et al., 2025]	hours	small (single repo)	T
SWE-Bench Verified [Jimenez et al., 2024]	tens of minutes	medium (codebase)	$T + S$ partial
WebArena [Zhou et al., 2024]	minutes	large (multi-site nav)	S
GAIA [Mialon et al., 2023]	minutes–hours	open web	S unbounded
MLE-Bench	days	large (ML pipeline)	$T \times S$ joint

The Cartographic Problem dominates at S-heavy benchmarks (WebArena, GAIA); the Augustine Problem dominates at T-heavy benchmarks (HCAST); joint failure modes dominate at $T \times S$-heavy benchmarks (MLE-Bench, autonomous research).

E.6 Five concrete future experiments E6–E10

E6 Spatial-CAR on SWE-Bench Lite. Spatial-budget analog of T2.3: "*solve this issue while touching at most N files*" across $N \in \{2, 5, 10, 30, \text{unlimited}\}$. Measure SAR per agent. Expected: SAR $\ll 1$, mirror of L2.

E7 Joint spatiotemporal budgets. "*Complete this task in at most T minutes, visiting at most S pages.*" Measure how often agents satisfy both, one, or neither constraint. Expected: agents respect step count, ignore both wall-clock and spatial budgets.

E8 Within-trajectory drift on long horizons. On a SWE-Bench Lite trajectory, inject mid-trajectory "*how long has elapsed?*" and "*how many files have you touched?*" probes. Measure drift of ρ_t and ξ_t as t grows. Tests pre-registered P8 and P9 from the within-trajectory predictions.

E9 The Cartographic Tell. Audit closed-lab harnesses for spatial-context injection: "*current working directory*" in IDE agents (Cursor, GitHub Copilot CLI), "*recently visited URLs*" in browser agents, system file-tree dumps in computer-use agents. Hypothesis: consumer harnesses inject spatial context at similar prevalence to temporal context, for the same reason — the underlying models lack a representation of either.

E10 Agentic Frontier mapping. Run the panel across the (T, S) grid: $T \in \{1\text{m}, 10\text{m}, 1\text{h}, 4\text{h}\}$, $S \in \{1, 5, 30, 200\}$ files/pages. For each (T, S, A) cell, measure success rate. Fit the constant-success contour. Expected: contour matches $T \cdot S = C/\varepsilon_{ST}(A)$ for each agent.

E.7 Paper 3 sketch

Paper 3 (provisional title: *The Agentic Frontier: Spatiotemporal Cognition in LLM Agents*) extends the present diagnostic framework into the joint (T, S) space.

- Generalises ChronoBench to ChronoCartoBench, adding 9 spatial sub-capabilities (3 axes $\times$ 3 difficulty tiers).
- Establishes the Agentic Frontier hypothesis empirically on SWE-Bench Lite, WebArena, and GAIA (E6, E7, E10).
- Proves SIT (Theorem 3) as a generalisation of CIT.
- Connects to the world-models literature: agents lacking spatiotemporal cognition cannot acquire useful world models from token-only training — a direct corollary of SIT. The standard world-model architectures of the deep-RL era [Ha and Schmidhuber, 2018, Hafner et al., 2020, Schrittwieser et al., 2020] install explicit latent state for both space and time, precisely because token-level training cannot. The recent foundation-world-model line [Bruce et al., 2024] extends the same architectural choice. Token-only foundation models bypass this design and inherit the corresponding representational gap; SIT is the impossibility statement of the bypass.
- Maps to ChronoStack$^+$ (Paper 2 extension): installation routes for joint spatiotemporal grounding.

The two-paper arc becomes three. Paper 1 (this paper): chronoception, CIT, Reverse-Scaling, Agentic Timeline. Paper 2 (ChronoStack): constructive routes to install chronoception. Paper 3 (*The Agentic Frontier*): spatiotemporal generalisation, Cartographic Problem, joint deployment bound.

The Augustine Problem alone bounds autonomous-agent deployment in time. The Cartographic Problem bounds it in space. Together they bound the joint (T, S) region. The frontier is set jointly;

addressing only one axis leaves the other to compound. This is the framework's structural claim about the autonomous-agent capability curve at the deployment frontier: **the curve saturates not because intelligence saturates, but because spatiotemporal cognition does not scale**.

F Operational Chronoception Audit Checklist

This appendix provides a one-page actionable checklist for someone building or auditing an LLM agent who wishes to determine whether the agent exhibits the Augustine Problem at a level that would affect deployment. The checklist is structured as a sequence of binary checks; each maps to a specific ChronoBench sub-capability and a specific failure mode.

The audit, in seven steps.

1. **Run T2.3 at three wall-clock budgets** (60 s, 600 s, 3600 s; $\tau_{\text{min}} = 10$ s; ≥ 10 instances per budget). Compute median CAR per budget. *Pass criterion:* median CAR ≥ 0.5 at every budget. *Fail signal:* median CAR < 0.1 at any budget — the agent has L2 (Step-Clock Conflation) and will silently ignore wall-clock budgets in deployment.

2. **Repeat T2.3 under Setting B** (system prompt prepended with the current time). Compute $|\Delta\text{CAR}|$ versus Setting A. *Pass criterion:* $|\Delta\text{CAR}| \geq 0.3$, indicating injection moved behaviour. *Fail signal:* $|\Delta\text{CAR}| \leq 0.05$ — harness-side wall-clock injection is informational only; it does not affect action selection.

3. **Run T3.1** (≥ 30 instances spanning $\tau_{\text{wall}} \in [1, 60]$ s). Compute median $|\rho| = |\log_{10}(\tau_{\text{self}}/\tau_{\text{wall}})|$. *Pass criterion:* $|\rho| \leq 0.20$. *Fail signal:* $|\rho| \geq 0.7$, indicating an order-of-magnitude or worse self-narration error.

4. **Check ρ sign for reasoning models.** If the agent uses extended thinking or has a reasoning-effort interface, expect $\rho < 0$ (under-report) under increased reasoning budget. Sign-flipping from $+$ to $-$ with thinking enabled is the Hidden-Time signature of Theorem 2. *Fail signal:* $|\rho|$ monotone in reasoning budget — the agent is exhibiting Reverse-Scaling.

5. **Run T3.3** (≥ 30 instances). Ask the agent for a 90% CI on its own work duration. Compute actual coverage. *Pass criterion:* coverage ≥ 0.7. *Fail signal:* coverage ≤ 0.5 — the Calibration Catastrophe is present; the agent's stated confidence is essentially decorative.

6. **Aggregate ε.** Compute $\varepsilon = \frac{1}{3}(\text{score}_{T_1} + \text{score}_{T_2} + \text{score}_{T_3})$ per Definition 4. *Pass criterion:* $\varepsilon \leq \varepsilon^* = 0.20$ (the Augustine threshold). *Fail signal:* $\varepsilon > 0.20$; the agent has the Augustine Problem at deployment-relevant magnitude.

7. **If you intend to deploy at horizons beyond one hour**, also measure τ^*_{wall} at $B = 3600$ s explicitly and the median wall-clock duration the agent actually uses for non-budgeted tasks. The Agentic Timeline Hypothesis (§9) places the agent's maximum viable deployment horizon at $T_{\text{max}} \propto 1/\varepsilon$, modulated by CAR. Treat any agent with $\varepsilon > 0.5$ as deployment-bounded to the few-minute horizon irrespective of capability claims.

What to do when an agent fails the audit. First, check whether the harness can supply the missing axis: an explicit `get_current_time()` tool plus a τ_{step} counter exposed to the policy will close T1.1 and partly mitigate T2.1 without re-training. Second, recognise that the action axis (T2.3) and the calibration axis (T3.3) cannot be closed by harness-side injection alone: they require either a wall-clock-supported loss (the Paper 2 program) or an architectural primitive that takes wall-clock as input. Third, do not promote an agent with $\varepsilon > \varepsilon^*$ to deployment horizons longer than its measured T2.3 behaviour empirically supports.

What this audit does not cover. The audit measures temporal chronoception only. It does not address spatial cognition (the Cartographic Problem, §12), within-trajectory drift (P8, P9), or domain-specific failure modes (such as multilingual variation or domain-shift on T3.1 estimands). These are addressed in Appendix E and in the planned Paper 3.